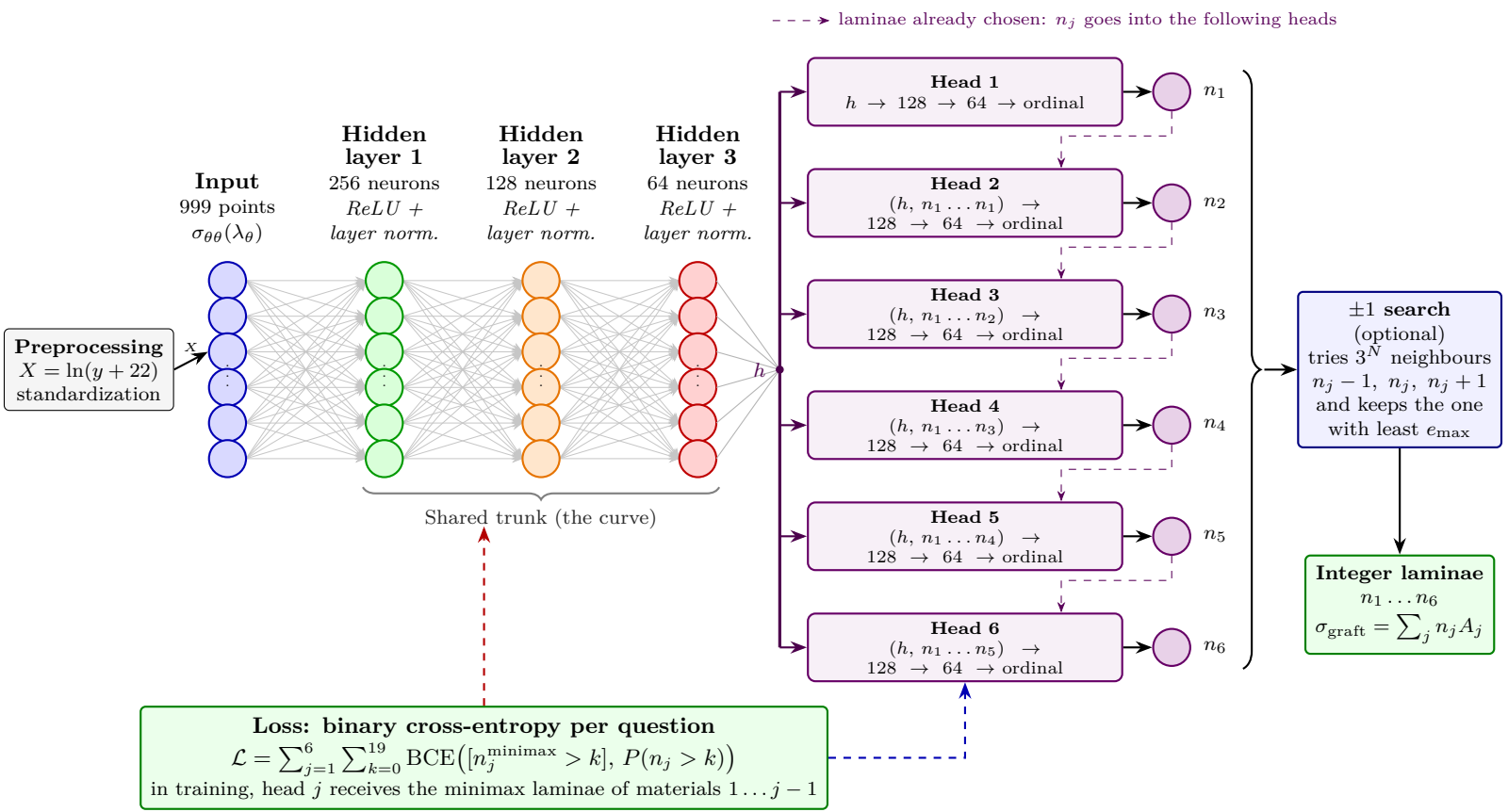
Ordinal chained network: shared trunk and one head per material

trunk $999 \to 256 \to 128 \to 64$; head j: $(h,\, n_1 \dots n_{j-1}) \to 128 \to 64 \to s_j$; 20 questions $P(n_j > k) = \sigma(s_j + b_{jk})$



Each head answers "more than k laminae?" for $k = 0 \dots 19$; n_j (0 to 20) is the number of "yes". It chooses knowing what the previous heads chose, so it can compensate between materials. It is used in order: material 1, then 2 knowing 1, and so on up to 6. About 400 000 parameters (6 materials).